RACE
Regular Analysis Through Continuous Exercise

## Fluid Mechanics : RACE-01

**1.** A submarine experiences a pressure of $5.05 \times 10^6$ *Pa* at a depth of $d_1$ in a sea. When it goes further to a depth of $d_2$, it experiences a pressure of $8.08 \times 10^6$ *Pa*. Then $d_2 - d_1$ is approximately (density of water = $10^3$ $kg/m^3$ and acceleration due to gravity = 10 $ms^{-2}$)

(A) 500 *m* (B) 400 *m* (C) 300 *m* (D) 600 *m*

**PFM193**

**2.** A hydraulic press can lift 100 *kg* when a mass '*m*' is placed on the smaller piston. It can lift _____*kg* when the diameter of the larger piston is increased by 4 times and that of the smaller piston is decreased by 4 times keeping the same mass '*m*' on the smaller piston.

**PFM194**

**3.** The pressure acting on a submarine is $3 \times 10^5$ *Pa* at a certain depth. If the depth is doubled, the percentage increase in the pressure acting on the submarine would be:

(Assume that atmospheric pressure is $1 \times 10^5$ *Pa* density of water is $10^3$ $kg\ m^{-3}$, $g = 10\ ms^{-2}$)

(A) $\frac{200}{3}$ % (B) $\frac{200}{5}$ % (C) $\frac{5}{200}$ % (D) $\frac{3}{200}$ %

**PFM195**

**4.** Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

**Assertion A:** When you squeeze one end of a tube to get toothpaste out from the other end, Pascal's principle is observed.

**Reason R:** A change in the pressure applied to an enclosed incompressible fluid is transmitted undiminished to every portion of the fluid and to the walls of its container.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) **A** is not correct but **R** is correct.

(B) **A** is correct but **R** is not correct.

(C) Both **A** and **R** are correct and R is the correct explanation of **A**.

(D) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**.

**PFM196**

**5.** Given below are two statements:

**Statement I :** Pressure in a reservoir of water is same at all points at the same level of water.

**Statement II :** The pressure applied to enclosed water is transmitted in all directions equally.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is false but Statement II is true.

(B) Both Statement I and Statement II are true.

(C) Statement I is true but Statement II is false.

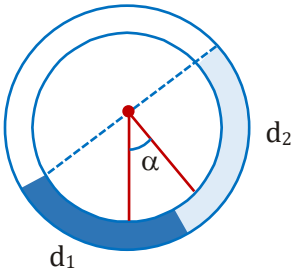
(D) Both Statement I and Statement II are false.

**PFM197**



For More Join: @IITJEE_Advanced

**6.** There is a circular tube in a vertical plane. Two liquids which do not mix and of densities $d1$ and $d2$ are filled in the tube. Each liquid subtends 90° angle at centre. Radius joining their interface makes an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is

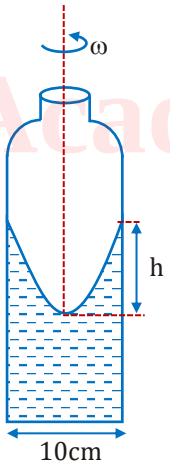



(A) $\frac{1+\tan\alpha}{1-\tan\alpha}$ (B) $\frac{1+\sin\alpha}{1-\cos\alpha}$s (C) $\frac{1+\sin\alpha}{1-\sin\alpha}$ (D) $\frac{1+\cos\alpha}{1-\cos\alpha}$

**PFM198**

**7.** A cylindrical vessel containing a liquid is rotated about its axis so that the liquid rises at its sides as shown in the figure. The radius of vessel is 5 *cm* and the angular speed of rotation is ω rad $s^{-1}$. The difference in the height, *h* (in *cm*) of liquid at the centre of vessel and at the side will be:

Rankers Academy JEE




(A) $\frac{25\omega^2}{2g}$ (B) $\frac{2\omega^2}{5g}$ (C) $\frac{5\omega^2}{2g}$ (D) $\frac{2\omega^2}{25g}$

**PFM199**

**8.** An open glass tube is immersed in mercury in such a way that a length of 8 *cm* extends above the mercury level. The open of the tube is then closed and sealed and the tube is raised vertically up by addition 46 *cm*. What will be length of the air column above mercury in the tube now ?

(Atmospheric pressure = 76 *cm* of *Hg*)

(A) 38 *cm* (B) 6 *cm* (C) 16 *cm* (D) 22 *cm*

**PFM200**



For More Join: @IITJEE_Advanced

**9.** A liquid of density is filled up to a height h in a cubical container of side length a. Torque, due to hydrostatic force acting on a wall, about the bottom edge of the wall is
(Assume atmospheric pressure is zero)

(A) $\frac{\rho g h^3 a}{2}$ (B) $\frac{\rho g h^3 a}{3}$ (C) $\frac{\rho g h^3 a}{6}$ (D) $\rho g h^3 a$

**PFM201**

**10.** If pressure at half the depth of a lake is equal to 2/ 3 pressure at the bottom of the lake, then what is the depth of the lake. (Pressure on the surface of lake = $10^5\ N/m^2$).

(A) 10 $m$ (B) 20 $m$ (C) 60 $m$ (D) 30 $m$

**PFM202**

Rankers Academy JEE

**ANSWER KEY**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer | C | 25600 | A | C | B | A | A | C | C | B |

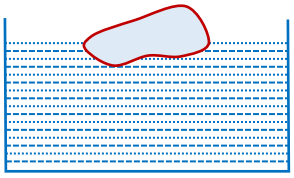


For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

## Fluid Mechanics : RACE-02

**1.** A body floats in a liquid contained in a beaker. The whole system as shown falls freely under gravity. The upthrust on the body due to liquid is



(A) Zero

(B) Equal to the weight of the liquid displaced

(C) Equal to the weight of the body in air

(D) Equal to the weight of the immersed portion of the body

**PFM203**

**2.** A small ball of relative density 0.8 falls into water from a height of $2m$. The depth to which the ball will sink is (Neglect viscous forces):

(A) 8 $m$ (B) 2 $m$ (C) 6 $m$ (D) 4 $m$

Rankers Academy JEE

**PFM204**

**3.** A cubical block of side 0.5 $m$ floats on water with 30% of its volume under water. What is the maximum weight that can be put on the block without fully submerging it under water?

(Take density of water = $10^3\ kg/m^3$)

(A) 65.4 $kg$ (B) 87.5 $kg$ (C) 30.1 $kg$ (D) 46.3 $kg$

**PFM205**

**4.** A hollow spherical shell at outer radius $R$ floats just submerged under the water surface. The inner radius of the shell is $r$. If the specific gravity of the shell material is $\frac{27}{8}$ w.r.t. water, the value of $r$ is:

(A) $\frac{4}{9}R$ (B) $\frac{8}{9}R$ (C) $\frac{1}{3}R$ (D) $\frac{2}{3}R$

**PFM206**

**5.** The velocity of a small ball of mass '$m$' and density $d_1$, when dropped in a container filled with glycerine, becomes constant after some time. If the density of glycerine is $d_2$, then the viscous force acting on the ball, will be :

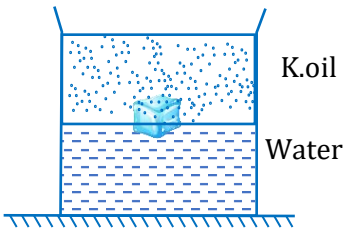
(A) $mg\left(1-\frac{d_1}{d_2}\right)$ (B) $mg\left(1-\frac{d_2}{d_1}\right)$ (C) $mg\left(\frac{d_1}{d_2}-1\right)$ (D) $mg\left(\frac{d_2}{d_1}-1\right)$

**PFM207**



For More Join: @IITJEE_Advanced

**6.** A cube of ice floats partly in water and partly in kerosine oil. Specific gravity of kerosine oil is 0.8 and ice is 0.9. What is the ratio of volume of ice in water to that in kerosine oil :-




(A) 1 : 1 (B) 1 : 8 (C) 8 : 1 (D) None of these

**PFM208**

**7.** A wooden block floating in a bucket of water has $\frac{4}{5}$ of its volume submerged. When certain amount of an oil is poured into the bucket, it is found that the block is just under the oil surface with half of its volume under water and half in oil. The density of oil relative to that of water is :-

(A) 0.5 (B) 0.7 (C) 0.6 (D) 0.8

**PFM209**

**8.** A boat carrying steel balls is floating on the surface of water in a tank. If the balls are thrown into the tank one by one, how will it affect the level of water

(A) It will remain unchanged (B) It will rise

(C) It will fall (D) First it will first rise and then fall

**PFM210**

Rankers Academy JEE

**9.** A 20 $N$ metal block is suspended by a spring balance. A beaker containing some water is placed on a weighing machine which reads 40 $N$. The spring balance is now lowered so that the block gets immersed in the water. The spring balance now reads 16 $N$. The reading of the weighing machine will be

(A) 36 $N$ (B) 60 $N$ (C) 44 $N$ (D) 56 $N$

**PFM211**

**10.** Consider a solid sphere of radius $R$ and mass density $\rho(r) = \rho_0\left(1 - \frac{r^2}{R^2}\right)$, $0 < r \le R$. The minimum density of a liquid in which it will float is :

(A) $\frac{\rho_0}{5}$ (B) $\frac{\rho_0}{3}$ (C) $\frac{2\rho_0}{3}$ (D) $\frac{2\rho_0}{5}$

**PFM212**

**ANSWER KEY**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer | A | A | B | B | B | A | C | C | C | D |



For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

## Fluid Mechanics : RACE-03

**1.** The top of a water tank is open to air and its water level is maintained. It is giving out 0.74 $m^3$ water per minute through a circular opening of 2 $cm$ radius in its wall. The depth of the centre of the opening from the level of water in the tank is close to :

(A) 9.6 $m$ (B) 4.8 $m$ (C) 2.9 $m$ (D) 6.0 $m$

**PFM213**

**2.** Water flows into a large tank with flat bottom at the rate of $10^{-4}\ m^3s^{-1}$. Water is also leaking out of a hole of area 1 $cm^2$ at its bottom. If the height of the water in the tank remains steady, then this height is:

(A) 4 $cm$ (B) 2.9 $cm$ (C) 1.7 $cm$ (D) 5.1 $cm$

**PFM214**

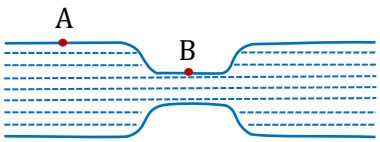
**3.** Water from a tap emerges vertically downwards with an initial speed of 1.0 $ms^{-1}$. The cross-sectional area of the tap is $10^{-4}\ m^2$. Assume that the pressure is constant throughout the stream of water and that the flow is streamlined. The cross-sectional area of the stream, 0.15 $m$ below the tap would be (Take $g$ = 10 $ms^{-2}$)

(A) $1 \times 10^{-5}\ m^2$ (B) $5 \times 10^{-5}\ m^2$ (C) $2 \times 10^{-5}\ m^2$ (D) $5 \times 10^{-4}\ m^2$

Pankers Academy JEE

**PFM215**

**4.** Water flows in a horizontal tube (see figure). The pressure of water changes by 700 $Nm^{-2}$ between $A$ and $B$ where the area of cross section are 40 $cm^2$ and 20 $cm^2$, respectively. Find the rate of flow of water through the tube. (Density of water = 1000 $kgm^{-3}$)




(Fig.)

(A) 1810 $cm^3/s$ (B) 3020 $cm^3/s$ (C) 2720 $cm^3/s$ (D) 2420 $cm^3/s$

**PFM216**

**5.** Two identical cylindrical vessels are kept on the ground, and each contain the same liquid of density $d$. The area of the base of both vessels is $S$ but the height of liquid in one vessel is $x_1$ and in the other, $x_2$. When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is :

(A) $gdS\,(x_2 + x_1)^2$ (B) $\frac{3}{4}gdS(x_2 - x_1)^2$ (C) $\frac{1}{4}gdS(x_2 - x_1)^2$ (D) $gdS(x_2^2 + x_1^2)$

**PFM217**



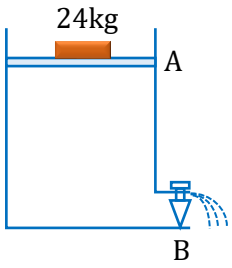
For More Join: @IITJEE_Advanced

**6.** A fluid is flowing through a horizontal pipe of varying cross-section, with speed $v\ ms^{-1}$ at a point where the pressure is $P$ Pascal. $P$ At another point where pressure is $\frac{P}{2}$ Pascal its speed is $V\ ms^{-1}$. If the density of the fluid is $\rho\ kg\ m^{-3}$ and the flow is streamline, then $V$ is equal to :

(A) $\sqrt{\frac{P}{2\rho}+v^2}$ (B) $\sqrt{\frac{P}{\rho}+v^2}$ (C) $\sqrt{\frac{2P}{\rho}+v^2}$ (D) $\sqrt{\frac{P}{\rho}+v}$

**PFM218**

**7.** Consider a water tank as shown in the figure. It's cross-sectional area is 0.4 $m^2$. The tank has an opening $B$ near the bottom whose cross-section area is 1 $cm^2$. A load of 24 $kg$ is applied on the water at the top when the height of the water level is 40 $cm$ above the bottom, the velocity of water coming out the opening $B$ is $v\ ms^{-1}$. The value of $v$, to the nearest integer, is___. [Take value of $g$ to be 10 $ms^{-2}$]



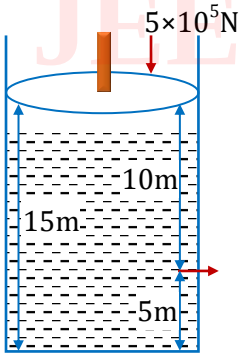

**PFM219**

**8.** The water is filled upto height of 12 $m$ in a tank having vertical sidewalls. A hole is made in one of the walls at a depth '$h$' below the water level. The value of '$h$' for which the emerging stream of water strikes the ground at the maximum range is _____ $m$.

**PFM220**

**9.** Consider a cylindrical tank of radius $1m$ is filled with water. The top surface of water is at 15 $m$ from the bottom of the cylinder. There is a hole on the wall of cylinder at a height of $5m$ from the bottom. A force of $5 \times 10^5\ N$ is applied an the top surface of water using a piston. The speed of efflux from the hole will be:
(Given atmospheric pressure $P_A = 1.01 \times 10^5\ Pa$, density of water $\rho_w = 1000\ kg/m^3$ and gravitational acceleration $g = 10\ m/s^2$)

(A) 11.6 $m/s$ (B) 10.8 $m/s$
(C) 17.8 $m/s$ (D) 14.4 $m/s$




**PFM221**

**10.** The surface of water in a water tank of cross section area 750 $cm^2$ on the top of a house is $h$ $m$. above the tap level. The speed of water coming out through the tap of cross section area 500 $mm^2$ is 30 $cm/s$. At that instant, $\frac{dh}{dt}$ is $x \times 10^{-3}\ m/s$. The value of $x$ will be_____.

**PFM222**

**ANSWER KEY**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer | B | D | B | C | C | B | 3 | 6 | C | 2 |



For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

## Fluid Mechanics : RACE-04

**1.** If '$M$' is the mass of water that rises in a capillary tube of radius '$r$', then mass of water which will rise in a capillary tube of radius '$2r$' is :

(A) $4M$ (B) $M$ (C) $2M$ (D) $\frac{M}{2}$

**PFM223**

**2.** A capillary tube made of glass of radius 0.15 $mm$ is dipped vertically in a beaker filled with methylene iodide (surface tension = 0.05 $Nm^{-1}$, density = 667 $kg\ m^{-3}$) which rises to height $h$ in the tube. It is observed that the two tangents drawn from liquid-glass interfaces (from opp. sides of the capillary) make an angle of 60° with one another. Then $h$ is close to ($g$ = 10 $ms^{-2}$).

(A) 0.137 $m$ (B) 0.172 $m$ (C) 0.087 $m$ (D) 0.049 $m$

**PFM224**

**3.** A solid sphere, of radius R acquires a terminal velocity $v_1$ when falling (due to gravity) through a viscous fluid having a coefficient of viscosity $\eta$. The sphere is broken into 27 identical solid spheres. If each of these spheres acquires a terminal velocity, $v_2$, when falling through the same fluid, the ratio ($v_1/v_2$) equals:

(A) 1/27 (B) 1/9 (C) 27 (D) 9

Rankers Academy JEE

**PFM225**

**4.** A air bubble of radius 1 $cm$ in water has an upward acceleration 9.8 $cm\ s^{-2}$. The density of water is 1 $gm\ cm^{-3}$ and water offers negligible drag force on the bubble. The mass of the bubble is ($g$ = 980 $cm/s^2$)

(A) 3.15 $gm$ (B) 4.51 $gm$ (C) 4.15 $gm$ (D) 1.52 $gm$

**PFM226**

**5.** In an experiment to verify Stokes law, a small spherical ball of radius $r$ and density $\rho$ falls under gravity through a distance $h$ in air before entering a tank of water. If the terminal velocity of the ball inside water is same as its velocity just before entering the water surface, then the value of $h$ is proportional to (ignore viscosity of air)

(A) $r$ (B) $r^4$ (C) $r^3$ (D) $r^2$

**PFM227**

**6.** When two soap bubbles of radii a and $b$ ($b > a$) coalesce, the radius of curvature of common surface is

(A) $\frac{ab}{b-a}$ (B) $\frac{a+b}{ab}$ (C) $\frac{b-a}{ab}$ (D) $\frac{ab}{a+b}$

**PFM228**

**7.** Two small drops of mercury each of radius $R$ coalesce to form a single large drop. The ratio of total surface energy before and after the change is :

(A) $2^{\frac{1}{3}} : 1$ (B) $1 : 2^{\frac{1}{3}}$ (C) 2 : 1 (D) 1 : 2

**PFM229**



For More Join: @IITJEE_Advanced

**8.** A raindrop with radius $R = 0.2\ mm$ falls from a cloud at a height $h = 2000\ m$ above the ground. Assume that the drop is spherical throughout its fall and the force of buoyance may be neglected, then the terminal speed attained by the raindrop is :
[Density of water $f_w = 1000\ kg\ m^{-3}$ and Density of air $f_a = 1.2\ kg\ m^{-3}$, $g = 10\ m/s^2$
Coefficient of viscosity of air = $1.8 \times 10^{-5}\ Nsm^{-2}$]
(A) 250.6 $ms^{-1}$ (B) 43.56 $ms^{-1}$ (C) 4.94 $ms^{-1}$ (D) 14.4 $ms^{-1}$

**PFM230**

**9.** Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).
**Assertion (A) :** Clothes containing oil or grease stains cannot be cleaned by water wash.
**Reason (R) :** Because the angle of contact between the oil/ grease and water is obtuse. In the light of the above statements, choose the correct answer from the option given below.
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is true but (R) is true.

**PFM231**

**10.** $1g$ of a liquid is converted to vapour at $3 \times 10^5\ Pa$ pressure. If 10% of the heat supplied is used for increasing the volume by 1600 $cm^3$ during this phase change, then the increase in internal energy in the process will be
(A) 4320 $J$ (B) 432000 $J$ (C) 4800 $J$ (D) $4.32 \times 10^8\ J$

**PFM232**

Rankers Academy JEE

**ANSWER KEY**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer | C | C | D | C | B | A | A | C | A | A |



For More Join: @IITJEE_Advanced